



GIET UNIVERSITY, GUNUPUR
SCHOOL OF ENGINEERING AND TECHNOLOGY
Department of CSE (AIML)
Database Management Systems (DBMS) — 3rd Semester
ASSIGNMENT — III

**Submission Date:- 08th
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Part A: Short Questions

1. Define normalization. What is its main purpose?
2. List any two anomalies caused by data redundancy, with a one-line example each.
3. Differentiate between full functional dependency and partial dependency.
4. Define transitive dependency with an example.
5. What is a lossless join decomposition?
6. State the condition for a binary decomposition of R into R1 and R2 to be lossless.
7. What is dependency preservation?
8. State the rule for a relation to be in 1NF & 2NF?
9. How does BCNF differ from 3NF?
10. What is a multivalued dependency (MVD)? Write its notation.
11. What is a join dependency? When is a relation in 5NF?
12. Give an example of a decomposition that is lossless but not dependency preserving.

Part B: Long Questions

1. Explain the need for normalization. Describe insertion, deletion and update anomalies with suitable examples.
2. Explain full, partial and transitive functional dependencies with examples for each.
3. Convert the following unnormalized table into 1NF and 2NF: STUDENT(Roll_No, Name, Course_ID, Course_Name, Marks), where Roll_No + Course_ID is the key and Course_ID → Course_Name.
4. Explain 3NF with an example. Normalize R(Emp_ID, Emp_Name, Dept_ID, Dept_Name) where Emp_ID → Dept_ID and Dept_ID → Dept_Name.
5. What is a multivalued dependency? Explain with the example COURSE(Course, Teacher, Book) and show the redundancy it causes.
6. Check whether the following relation is in 4NF. If not, decompose it and show that the original table can be recovered by a join. Assume Employee →→ Skill and Employee →→ Language.

Employee	Skill	Language
E1	Java	English
E1	Java	Hindi
E1	SQL	English
E1	SQL	Hindi
E2	Python	English

1. Hospital case study: A table is given as PATIENT_VISIT(Patient_ID, Patient_Name, Patient_Phone, Doctor_ID, Doctor_Name, Specialization, Dept_ID, Dept_Name, Visit_Date, Diagnosis, Fee). Assume: a patient has one phone number; a doctor belongs to one department and has one specialization; the fee depends on Doctor_ID. A patient can visit the same doctor on different dates.
 - (a) Write all functional dependencies.
 - (b) Identify the candidate key.
 - (c) List the anomalies present.
 - (d) Normalize step by step to 1NF, 2NF, 3NF and BCNF, showing the resulting relations and keys.